



Department of Information Technology
Academic Year 2025 - 2026 (Odd Semester)

Degree, Semester & Branch: III Semester B.Tech. IT

Course Code & Title: CS3352 & Foundations of Data Science

Name of the Faculty member (s): Dr.V. Anusuya, ASCP/IT

Innovative Practice Description

- **Unit / Topic:** Unit II / Describing data with Averages, Describing Variability
- **Course Outcome:** CO2
- **Topic Learning Outcome:** TLO4
- **Activity Chosen:** Think Pair Share
- **Justification:**

In unit II, describing data with averages and Describing Variability is an important concept in statistics and the questions were asked in most of the university questions. This activity helps the students to recall and practice the concepts.

- **Time Allotted for the Activity:** 10 Minutes
- **Details of the Implementation:**
 - The topics describing data with averages, describing Variability taught in the previous classes. The time allotted for the activity is 10 Minutes.
 - For each pair students one question was prepared and assign to each pair as shown in Figure 1a and Figure 1b.
 - The pairs are discussed and they write down the answers.
 - Asked each pair to present their solution Mr.M.Karthik and Sudharshan S M team, Iptika and Ram Selvalakshmi team as shown in Figure 2a and Figure 2b.
 - The answers for the questions as shown in Figure 3

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO2	3	3	2	2	2	1	2	2	3	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO2	3	3	2	2	2	1	2	2	3	2
Justification for correlation	Apply basic Knowledge and fundamentals of mathematics for data statistics	The students will be able to identify the need for analysis of complex engineering problems	The students will be able to design the process flow for describing data.	The students will be able to analyze data and represent data to provide valid conclusions	The students will be able to functional individually or as a team in analyzing, representing data and perform statistical techniques	The students will be able to interpret statistical descriptions of data and prepare as a report or presentation	Ability to recognize the need for statistical techniques for describing data	The students able to exhibit the knowledge in statistical descriptions of data for application development.	The students can Exhibit knowledge in statistical techniques of data in the field of IoT, Cloud Computing to develop IT solutions	The students able to exhibit knowledge statistical techniques of data to develop a quality product.

• **Images / Screenshot of the practice:**

Question:

1. a) Calculate the standard deviation and mean of a dataset with values: [5, 8, 12, 15, 20].
b) Find the mode of the following numbers: 5, 8, 6, 9, 8, 3, 5, 2, 8.
2. a) Calculate the median for the given grouped data: [10-20]: 5, [20-30]: 12, [30-40]: 8, [40-50]: 6.
b) Calculate the mode of the following test scores: 88, 92, 78, 92, 85, 90, 88, 93.
3. a) . In a survey, participants' ages are recorded. Find the median age from the following ages: 29, 31, 35, 40, 41, 45, 49.
b) . Find the standard deviation for a dataset with values: [10, 10, 10, 10, 10].
4. a) . Find the median for the following dataset: [0-10]: 15, [10-20]: 25, [20-30]: 10.
b) Calculate the mode of the following test scores: 88, 92, 78, 92, 85, 90, 88, 93.
5. a) . Determine the standard deviation for a dataset with values: [2, 4, 6, 8, 10].
b) In a survey, participants' ages are recorded. Find the median age, mean from the following ages: 29, 31, 35, 40, 41, 45, 49.
6. a) What is the mode of the shoe sizes for a group of 20 people: 8, 9, 10, 9, 7, 8, 9, 10?
b) Determine the median for grouped data: [0-5]: 8, [5-10]: 12, [10-15]: 20, [15-20]: 15.
7. a) Calculate the standard deviation of a sample with values: [12, 14, 18, 20, 22].
b) Calculate the mean, median for the following dataset of annual incomes: \$45,000, \$50,000, \$55,000, \$60,000, \$75,000, \$80,000, \$100,000.
8. a) Find the median for grouped data: [0-100]: 7, [100-200]: 18, [200-300]: 11.
b) Calculate the mode of the temperatures recorded for a week: 78°F, 82°F, 78°F, 77°F, 83°F, 79°F, 78°F.
9. a) . Determine the median for the grouped data: [0-25]: 3, [25-50]: 9, [50-75]: 16.
b) . Find the mode of the following series: 12, 18, 24, 12, 30, 18, 36, 24.
10. a) Find the standard deviation for a dataset with values: [3, 7, 11, 15, 19].
b) . Find the median height of a group of individuals with the heights (in inches): 62, 64, 66, 68, 70, 72, 74.
11. a) A data set represents the number of hours worked by employees in a week: 32, 40, 42, 45, 50, 55. Calculate the median.
b) Calculate the population standard deviation for a dataset with values: [8, 16, 24, 32, 40].
12. a) Determine the median for the grouped data: [0-25]: 3, [25-50]: 9, [50-75]: 16.
b) . Determine the mode of the following list of book prices: \$12.50, \$15.75, \$12.50, \$18.25, \$20.50, \$15.75.
13. a) Find the median height of a group of individuals with the heights (in inches): 62, 64, 66, 68, 70, 72, 74.
b) Determine the standard deviation for a dataset with values: [5, 12, 19, 26, 33].
14. a) A data set represents the number of hours worked by employees in a week: 32, 40, 42, 45, 50, 55. Calculate the median.
b) Calculate the standard deviation of a dataset with values: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10].
15. a) . Calculate the median for the dataset grouped as [0-5]: 12, [5-10]: 14, [10-15]: 8, [15-20]: 11.
b) . Calculate the mode of the number of siblings for a group of 50 students: 1, 2, 2, 0, 1, 3, 1, 0.
16. a) Find the standard deviation for a dataset with values: [0, 5, 10, 15, 20, 25, 30, 35, 40, 45].
b) From a set of daily website visit counts: 200, 250, 300, 350, 400, 450, 600, find the median.
17. a) . Find the median for the grouped data: [0-30]: 20, [30-60]: 15, [60-90]: 10.
b) Find the mode of the ages of a group of people: 35, 27, 41, 35, 29, 29, 41, 35, 38.
18. a) . Determine the median for the dataset grouped as [0-40]: 6, [40-80]: 12, [80-120]: 18.
b) Determine the mode of the number of daily website visitors over a week: 100, 150, 125, 150, 100, 175, 200.
19. a) Calculate the standard deviation of a sample with values: [22, 24, 25, 28, 30, 32].
b) Calculate the mean, median for a dataset representing the response times (in seconds) for a web application: 3.2, 4.5, 5.1, 5.8, 6.4, 7.0.

20. a) . Calculate the mode of the monthly rainfall measurements (in inches) for a year: 2.5, 3.0, 1.5, 2.5, 1.0, 3.5, 2.5, 1.5, 3.0.
 b) . Determine the standard deviation for a dataset with values: [2, 2, 2, 8, 8, 8].
21. a) Calculate the median for the following grouped data: [0-10]: 7, [10-20]: 14, [20-30]: 5.
 b) . Find the mode of the following numbers: 4, 5, 6, 6, 8, 8, 8, 10, 11, 12.
22. a) . Calculate the population standard deviation for a dataset with values: [3, 3, 3, 3, 3].
 b) . Calculate the median for a dataset of product prices (in dollars): \$15, \$20, \$25, \$30, \$35, \$40, \$50.
23. a) . Find the median for the grouped data: [0-50]: 8, [50-100]: 10, [100-150]: 12.
 b) Determine the mode of the following test scores: 88, 92, 78, 92, 85, 90, 88, 93, 88, 85.
24. a) . Determine the standard deviation for a dataset with values: [10, 20, 30, 40, 50].
 b) What is the mode of the shoe sizes for a group of 30 people: 8, 9, 10, 9, 7, 8, 9, 10, 11, 10?
25. a) . Determine the mean, median for grouped data: [0-20]: 5, [20-40]: 8, [40-60]: 10, [60-80]: 7.
 b) Calculate the mode of the temperatures recorded for a month: 78°F, 82°F, 78°F, 77°F, 83°F, 79°F, 78°F, 85°F.
26. a) Determine the median of a dataset representing the sizes (in square feet) of houses in a neighborhood: 1500, 1600, 1700, 1800, 1900, 2000.
 b) Find the standard deviation for a dataset with values: [1, 1, 2, 2, 3, 3, 4, 4, 5, 5].
27. a) . Find the median of a dataset representing the weights (in pounds) of a group of individuals: 140, 155, 165, 175, 185, 200.
 b) . Calculate the standard deviation of a sample with values: [17, 19, 21, 24, 26, 28].
28. a) Find the mode of the following series: 12, 18, 24, 12, 30, 18, 36, 24, 18, 36.
 b) Calculate the median for the dataset grouped as [0-25]: 15, [25-50]: 20, [50-75]: 10.
29. a) . Determine the median for the grouped data: [0-15]: 12, [15-30]: 8, [30-45]: 10, [45-60]: 14.
 b) Determine the mode of the following list of book prices: \$12.50, \$15.75, \$12.50, \$18.25, \$20.50, \$15.75, \$12.50, \$18.25.
30. a) . Find the median of a dataset representing the weights (in pounds) of a group of individuals: 14, 15, 16, 17, 18, 20.
 b) . Calculate the standard deviation of a sample with values: [17, 19, 21, 24, 26, 28].
31. a) Find the mode of the following series: 123, 184, 245, 124, 300, 181, 364, 241, 181, 365.
 b) Calculate the median for the dataset grouped as [0-25]: 15, [25-50]: 20, [50-75]: 10.

Figure1a: Questions given to each pair



Figure1b: Questions given to each pair



Figure 2a: Presentation by Iptika, Ram selvalakshmi



Figure 2b Students during the activity

14(a) A data set represents the no. of hours workers by employees in a week: 32, 40, 42, 45, 50, 55. calc. the median

(a) 32, 40, 42, 45, 50, 55

$$= \frac{1}{2} [(n/2)^{\text{th}} \text{ term} + \{(n/2+1)^{\text{th}} \text{ term}\}]$$

$$= \frac{1}{2} [(6/2)^{\text{th}} \text{ term} + (3+1)^{\text{th}} \text{ term}]$$

$$= \frac{1}{2} [3^{\text{rd}} \text{ term} + 4^{\text{th}} \text{ term}]$$

$$= \frac{1}{2} [42 + 45]$$

$$= 87/2 = 43.5$$

(b) Calculate the standard deviation of a data set with values: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10].

$$\text{Mean} = \frac{1+2+3+4+5+6+7+8+9+10}{10}$$

$$= \frac{55}{10} = 5.5$$

$$\sigma^2 = \frac{\sum (x - \mu)^2}{N}$$

$$= \frac{(1-5.5)^2 + (2-5.5)^2 + (3-5.5)^2 + (4-5.5)^2 + (5-5.5)^2 + (6-5.5)^2 + (7-5.5)^2 + (8-5.5)^2 + (9-5.5)^2 + (10-5.5)^2}{10}$$

$$= \frac{82.5}{10}$$

$$\sigma^2 = 8.25$$

$$\sigma = \sqrt{8.25} = 2.87$$

6) a) What is the mode of the shoe sizes for a group of 20 people: 8, 9, 10, 9, 7, 8, 9, 10?

b) Determine the median for grouped data: [0-5]: 8, [5-10]: 12, [10-15]: 20, [15-20]: 15.

6) a) Ans:

Mode of shoe sizes of group of 20 people is 9.

6) b) Ans:

Range	Mid value (x _i)	value (f _i)	x _i f _i
0-5	2.5	8	20
5-10	7.5	12	90
10-15	12.5	20	250
15-20	17.5	15	262.5
	$\Sigma x_i = 38$		$\Sigma x_i f_i = 595$

Median = $\frac{\Sigma x_i f_i}{\Sigma x_i} = \frac{595}{38} = 15.66$

Median = 15.66

Figure 3 Answer for the Questions

Reflective Critique:

❖ Feedback of practice from students and other stakeholders:

- The activity helped in better understanding of concepts such as *mean, median, mode, range, variance, and standard deviation*.
- Many students felt more confident in solving numerical problems after discussing with partners.
- Students found the *Think-Pair-Share* strategy engaging because it allowed them to first reflect individually, then discuss with peers, and finally share with the class.

❖ Benefit of the practice:

- Students gained clarity on statistical measures like *mean, median, mode, range, variance, and standard deviation* through peer discussion and explanation.

- Discussing answers first in pairs reduced anxiety and built confidence to share ideas with the larger group.
- The activity encouraged all students to participate rather than passively listening, leading to higher attention and interest in the topic.
- The activity encouraged active participation from a majority of students, even those who are usually hesitant to speak in class.

❖ ***Challenges faced in implementation:***

- Some pairs needed more time to calculate and interpret measures like variance and standard deviation, which extended beyond the planned duration.
- Not all pairs/groups could present their answers in the “share” phase due to class time constraints.

References:

- ❖ https://www.ritrjpm.ac.in/images/B.Tech-IT/2024-2025/VA_CS3452_Think_Pair_Share.pdf.
- ❖ https://www.ritrjpm.ac.in/images/B.Tech-IT/2024-2025/VA_CS3352_Think_Pair_Share.pdf

Signature of Faculty Member

HOD